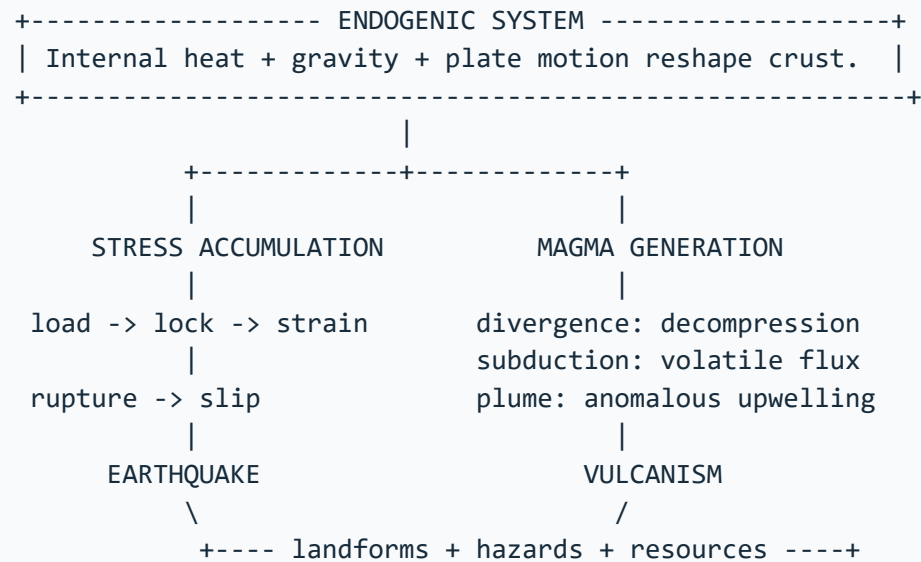


ASCII MASTER FLOW – PANEL 1/12: Endogenic system: plate motion, strain release and magma generation



FAULT PAIRS

tension -> normal | compression -> reverse/thrust | shear -> strike-slip

Plate motion loads a region; sudden fault rupture is the immediate quake source.

MUST REMEMBER: Vulcanism and seismicity follow magma generation, ascent, eruption style, fault rupture, wave propagation and site response; India requires Himalayan collision, Kachchh intraplate strain, Peninsular reservoirs/faults, Andaman subduction and Barren Island anchors.

ASCII MASTER FLOW – PANEL 2/12: Earthquake source geometry, depth and seismogram logic

Diagram illustrating the relationship between station A, station B, epicentre, focus / hypocentre, and rupture begins on fault plane.

The diagram shows a horizontal line representing the surface. Above the line, the word "SURFACE" is centered. Below the line, the word "epicentre" is centered. To the left of the epicentre is "station A" and to the right is "station B". Dashed lines connect station A to the epicentre and the epicentre to station B. Below the epicentre, a vertical line descends to the word "focus / hypocentre". Below "focus / hypocentre", the text "rupture begins on fault plane" is centered.

SEISMOGRAM ARRIVAL ORDER

P first -> S second -> Love and Rayleigh later

|
P-S interval at one station gives source distance

three or more distance circles estimate the epicentre

DEPTH CLASS

shallow: 0-70 km | intermediate: 70-300 km | deep: 300-700 km

SEQUENCE FIREWALL

foreshock, mainshock and aftershock are relational labels;
a swarm has no single clearly dominant event.

Epicentre is a vertical projection, not necessarily the worst-damaged place.

ASCII MASTER FLOW – PANEL 3/12: Seismic waves, shadow zones, magnitude and intensity

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion

P shadow about 103-142 deg | direct S shadow beyond about 103 deg

MAGNITUDE: one source measure; Mw uses seismic moment

moment = rigidity x fault area x average slip

+1 local magnitude gives about 10x amplitude and 31.6x energy

INTENSITY: place-specific observed effects; many values for one event

source x path x site x exposure x vulnerability / capacity -> damage

ASCII MASTER FLOW – PANEL 4/12: Global seismic belts, cascading hazards and tsunami sequence

GLOBAL EARTHQUAKE DISTRIBUTION

```
+-----+-----+-----+
| Circum-Pacific      | Mediterranean-   | Ridges/transforms |
| trench + slab + arc | Himalayan collision | shallow events     |
| megathrust + tsunami | complex convergence | basaltic ridges    |
+-----+-----+-----+
Intraplate events reactivate inherited structures; stable is not aseismic.
```

CASCADE

```
shaking -> amplification / liquefaction / lateral spreading
         -> landslide / fire / lifeline failure / dam distress
```

TSUNAMI

```
submarine megathrust -> vertical sea-floor displacement
-> long low deep-ocean wave -> coastal shoaling
-> run-up + inundation + return flow
```

First wave need not be largest; withdrawal need not occur before every event.
Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.

ASCII MASTER FLOW – PANEL 5/12: Volcano anatomy, magma controls, landforms and eruption styles

```
      ash plume
        /\
       /  \ crater
lava  --    -- lava
      |vent|
dyke  ----|  |---- sill
      |conduit|
      magma chamber
```

Magma below surface | lava after eruption

Silica up -> viscosity up -> degassing harder -> explosivity may rise

Temperature up -> viscosity generally down

Retained gas + decompression -> bubbles -> fragmentation

FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill

Dyke cuts layers; sill follows layers.

STYLE: fissure/Icelandic -> Hawaiian -> Strombolian -> Vulcanian -> Plinian

FORM: shield | composite | cinder cone | dome | fissure plateau

STATUS: active | dormant | extinct; status is not a compulsory life cycle.

ASCII MASTER FLOW – PANEL 6/12: Volcanic settings, global belts, products and environmental effects

+-----+	+-----+	+-----+
SETTING	MELTING MECHANISM	EXPRESSION
+-----+	+-----+	+-----+
Divergent	decompression	ridge, rift, basalt
Subduction	slab volatile flux	trench-slab-arc system
Plume	mantle upwelling	flood basalt, hot spot
+-----+	+-----+	+-----+

Ring of Fire = trench + descending slab + earthquakes + volcanic arc
Plume model: broad head may form flood basalt; tail may form age chain.
Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

HAZARDS: lava | ash | density currents | lahars | gases
CLIMATE: stratospheric sulphate aerosol can cause temporary cooling
BENEFITS: new crust/land | soil parent | geothermal energy | minerals

Impact converges through wind, relief, drainage, exposure and preparedness.

ASCII MASTER FLOW – PANEL 7/12: India seismic geography, zonation, institutions and resilience stack

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors
North-East : convergence, strike-slip complexity
Andaman : subduction, tsunami, Barren Island volcanism
Kachchh : inherited rift and fault reactivation
Peninsula : relatively stable; Koyna and Latur show residual risk
Indo-Gangetic : alluvium, amplification and high exposure

|

MACROZONATION: broad design hazard, not prediction

MICROZONATION: geology + soil + motion + water + liquefaction + GIS

|

NCS/MoES -> monitoring | GSI -> geology | BIS -> standards

NDMA/NIDM -> guidance | INCOIS -> tsunami warning

States/ULBs -> approvals + inspection + enforcement

|

site -> regular load path -> ductility -> connections

-> non-structural safety -> quality control -> retrofit -> drills

Early warning detects rupture after it starts; it is not prediction.

ASCII MASTER FLOW – PANEL 8/12: Hazard-risk-response and Mains answer spine for seismic India

QUESTION: Why can similar earthquakes produce unequal disasters?

HAZARD: source size + depth + rupture geometry + path

SITE: basin sediment + groundwater + slope + liquefaction potential

EXPOSURE: people + buildings + lifelines + time of occurrence

VULNERABILITY: weak design + irregular form + poor maintenance

CAPACITY: codes + enforcement + retrofit + warning + response

RISK VERDICT: hazard becomes disaster through social and spatial filters

15-MARK INDIA SPINE

tectonic regions -> site modifiers -> exposure/vulnerability -> cases

-> zonation and microzonation -> construction and governance -> conclusion

TRAPS: focus is not epicentre | magnitude is not intensity

tsunami is not a tide | caldera is not an ordinary crater

zone map is not prediction | code publication is not enforcement.

ASCII MASTER FLOW – PANEL 9/12: Evidence ladder and confidence control

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
 2. Extinct volcano Geological evidence indicates eruption is no longer...
 3. geophysical evidence; a long-quiescent volcano may still be dormant.
- VERDICT: Source type controls the strength and limits of the historical claim.

ASCII MASTER FLOW – PANEL 10/12: Examiner traps and contested boundaries

CLOSE DISTINCTIONS

1. Causal method Initial condition -> energy/force or driver ->...
2. PIB/MoES - updated seismic zonation map statement, 17 December 2025,...
3. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT

VERDICT: Qualification earns marks; false certainty is a hard failure.

CLOSE DISTINCTION: Magnitude measures source size whereas intensity records effects; focus differs from epicentre, P differs from S and surface waves, and a tsunami requires water-column displacement rather than every submarine earthquake.

ASCII MASTER FLOW – PANEL 11/12: Integrated answer spine and qualified conclusion

ANSWER SPINE

1. Cross-owned but directly relevant Mains PYQ 1
2. Original solved Mains practice 6
3. Practice contract Every solved item has demand decoding, a detailed...

VERDICT: Claim, named evidence, analysis and qualification lead to a graded...

EVIDENCE LIMIT: Hazard is not disaster: exposure, vulnerability, construction and preparedness
mediate loss; seismic zonation is probabilistic and revised through evidence, while
prediction claims must not be confused with monitoring or early warning.

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
 2. Extinct volcano Geological evidence indicates eruption is no longer...
 3. geophysical evidence; a long-quiescent volcano may still be dormant.
- VERDICT: Source type controls the strength and limits of the historical claim.